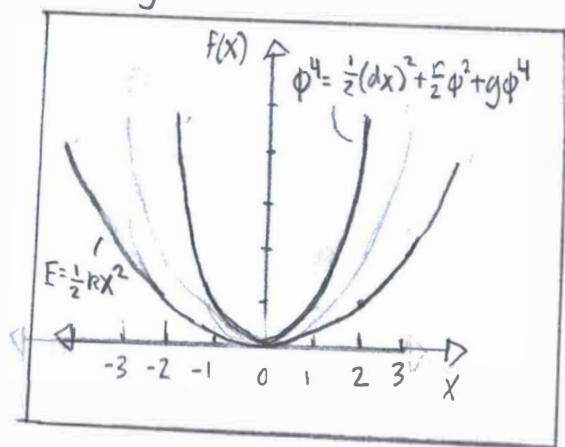


Chapter 4: Perturbation Theory:

Exercise #1:
(pg 175)



" ϕ^4 - theory - a prototypical Harmonic Oscillator"

(pg 175) "Action Expansion"

$$S[\phi] = \sum_k \left(\phi_k (c_1 + c_2 k \cdot k) \phi_{-k} + c_3 \phi_k h_{-k} \right) + \frac{c_4}{N} \sum_{k_1, \dots, k_4} \phi_{k_1} \phi_{k_2} \phi_{k_3} \phi_{k_4} \delta_{k_1+k_2+k_3+k_4} + O(k^4)$$

$$\text{where } \phi_i = \frac{1}{\sqrt{N}} \sum_k e^{-ik \cdot r_i} \phi(k)$$

$$K_{ij} = \frac{1}{N} \sum_k e^{-ik(r_i - r_j)} K(k)$$

(pg 175) " ϕ^4 - theorem"

$$S[\phi] = - \sum_{i,j} \phi_i K_{ij} \phi_j + \sum_i \phi_i h_i + \sum_i \ln \cosh(2 \sum_j K_{ij} \phi_j)$$

Term-by-term comparison:

$$\textcircled{1} \quad \phi_k (c_1 + c_2 k \cdot k) \phi_{-k} = \phi_i K_{ij} \phi_j$$

$$\textcircled{2} \quad c_3 \phi_i h_i = \phi_i h_i$$

$$\textcircled{3} \quad \frac{c_4}{N} \phi_{k_1} \phi_{k_2} \phi_{k_3} \phi_{k_4} \delta_{k_1+k_2+k_3+k_4} = \ln \cosh(2 \sum_j K_{ij} \phi_j)$$

(From problem) "Expansion"

$$K_{ij} = K(0) \phi(k) + \frac{1}{2} k^2 K''(0) \cdot \phi(k)$$

$$① C_1 = k(0) \phi(k)$$

$$② C_2 = \frac{1}{2} k^2 k''(0) \phi(k)$$

$$③ C_3 = -1$$

$$④ C_4 = 4k(0)\phi(k) - \frac{4}{3}k(0)\phi(k)$$

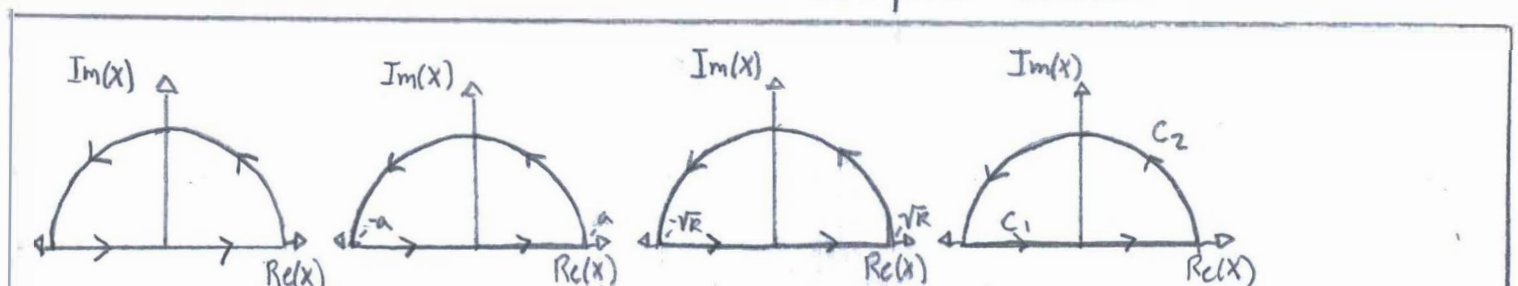
Log (cosh) Series:

$$\ln(\cosh(x)) = \frac{x^2}{2} - \frac{x^4}{12}$$

Exercise #2:
(pg 177)

The correlation function is an expectation to amplitude for fields.

"Complex Contour"



$$\oint F(x) dx = \int_{-a}^a F(x) dx = \int_{-\infty}^{\infty} \frac{e^{ikx}}{k^2+1} dx = \oint_{C_1} F(x) dx + \oint_{C_2} F(x) dx = 2\pi i \sum \text{Res}(F, x)$$

"Residue Theorem"

Cite: Cauchy, A.

"Mémoire sur les intégrales
définies, prises entre des
limites imaginaires"

Paris, France (1825)

Exercise #3:
(pg 178)

(4.12) "Green function-inverse Transform"

$$G_0(x) = \frac{1}{L^d} \sum_p e^{-ipx} G_{0,p}$$

$$\approx \int \frac{d^d p}{(2\pi)^d} \frac{e^{-ipx}}{p^2 + r}$$

(4.17) "For dimensions $d > 1$, the integral is divergent"

$$G_0(x) = \int \frac{d^d k}{(2\pi)^d} \frac{1}{p^2 + r}$$

Note: The exercise encounters a critical dimension ($d > 1$).

Two-dimensions ($d=2$):

① Complex integral:

$$G_0(x) \stackrel{d=2}{=} \int_{-\infty}^{\infty} \frac{d^d k}{(2\pi)^2} \frac{e^{-ikx}}{k^2 + r}$$

② Vertical asymptotes:

$$= \int_{-\infty}^{\infty} \frac{d^2 k}{(2\pi)^2} \frac{e^{-ikx}}{(k - ir^{1/2})(k + ir^{1/2})}$$

asymptotes at $k = \pm i\sqrt{r}$

③ Residue to Complex Contour:

$$= \frac{1}{(2\pi)^2} \int_{-\infty}^{\infty} dk \left[2\pi i \operatorname{Res}_{k=\sqrt{-r}} \left[\int dk \frac{e^{-ikx}}{k^2 - ir^{1/2}} \right] \right. \\ \left. + 2\pi i \operatorname{Res}_{k=-\sqrt{-r}} \left[\int dk \frac{e^{-ikx}}{k^2 + ir^{1/2}} \right] \right]$$

④ Evaluate one root:

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} dk i \cdot \left[\frac{e^{-\sqrt{r}x}}{2i\sqrt{r}} \right]$$

$$= \frac{1}{4\pi} \int_{-\infty}^{\infty} dk \frac{e^{-\sqrt{r}x}}{\sqrt{r}}$$

$$\approx \begin{cases} \frac{1}{2\pi} \ln \frac{\sqrt{r}x}{2} & x \ll 1/\sqrt{r} \\ \frac{1}{2} (2\pi\sqrt{r}|x|)^{-1/2} e^{-\sqrt{r}x} & x \gg 1/\sqrt{r} \end{cases}$$

Three-dimensions (d=3):

$$G_0(x) \stackrel{d=3}{=} \int \frac{d^3k}{(2\pi)^3} \frac{e^{-ikx}}{k^2 + r}$$

$$= \frac{1}{8\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} d^2k \frac{e^{-\sqrt{r}x}}{\sqrt{r}}$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{d^2k}{4\pi} \frac{e^{-\sqrt{r}x}}{\sqrt{r}}$$

Book form, but divergent.

Exercise #4:
(pg 180)

(3.21) "Product over infinite sets"

$$\langle v(x_1) v(x_2) \dots v(x_{2n}) \rangle = \sum_{\substack{\text{pairs} \\ \{x_1, \dots, x_{2n}\}}} A^{-1}(x_{k_1}, x_{k_2}) \dots A^{-1}(x_{k_{2n-1}}, x_{k_{2n}})$$

$$\langle \phi(x) \phi(y)^4 \phi(y')^4 \phi(x') \rangle_0 = 2!^{10-1} \text{ total } x(10-1) \text{ permutations} \\ = 945 \text{ terms}$$

Note: Incorrect typeset in line (4.18)

$$\begin{aligned} \langle \phi(x) \phi(y)^4 \phi(y')^4 \phi(x') \rangle &= 9 G_0(x-x') G_0(0)^4 \\ &+ 72 G_0(x-x') G_0(y-y') G_0(0)^2 + 24 G_0(x-x') G_0(y-y')^4 \\ &+ 72 G_0(x-y) G_0(x'-y') G_0(0)^3 \\ &+ 288 G_0(x-y) G_0(x'-y) G(y-y') \\ &+ 192 G_0(x-y) G(y-y') G(x'-y') \\ &+ 288 G(x-y) G(y-y') G(x'-y') \end{aligned}$$

Exercise #5:
(pg 181)

The problem uses 

A connection ( — ) describes a 2-point correlation, with an integral.

Diagram #1: $9 G(x-x') G_0(0)^4$



Diagram #2: $72 G(x-x') \circ G(y-y') \circ G_0(0)^2$



Diagram #3: $24 G_0(x-x') G_0(y-y')^4$

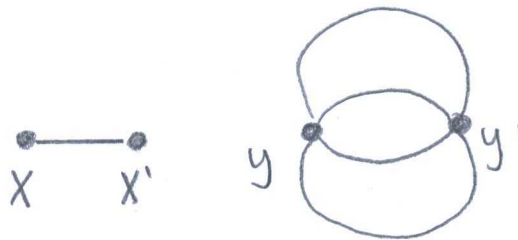


Diagram #4: $72 G_0(x-y) G_0(x'-y') G_0(0)^3$

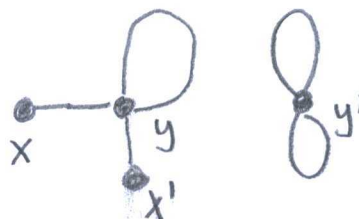


Diagram #5: $288 G(x-y) G_0(y-y') G_0(x'-y)$

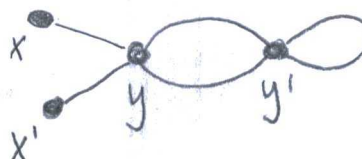


Diagram #6: $192 G_0(x-y)G_0(y-y')G_0(x'-y)$

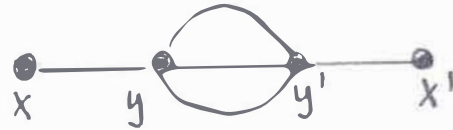
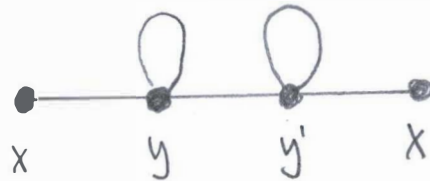


Diagram #7: $288 G(x-y)G(y-y')G(x'-y')$



The correlation function is
on line (4.15).

Exercise #6:
(pg 134)

(4.14) "Functional expectation value"

$$\langle X[\phi] \rangle \simeq \frac{\sum_{n=0}^{\infty} \frac{(-g)^n}{n!} \langle X[\phi] (\int d^d x \phi^4)^n \rangle_0}{\sum_{n=0}^{\infty} \frac{(-g)^n}{n!} \langle (\int d^d x \phi^4)^n \rangle_0}$$

$$\simeq \sum_{n=0}^{n_{max}} X^{(n)}$$

$$\langle X[\phi] \rangle \simeq \frac{\sum_{n=0}^2 \frac{(-g)^n}{n!} \langle X[\phi] (\int d^d x \phi^4)^n \rangle_0}{\sum_{n=0}^2 \frac{(-g)^n}{n!} \langle (\int d^d x \phi^4)^n \rangle_0}$$

$$= \frac{N_0 + N_1 + \frac{1}{2} N_2}{1 + D_1 + \frac{1}{2} D_2}$$

Binomial Expansion:

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2$$

$$\approx N_0 - N_1 - N_0 D_1 + \frac{1}{2} N_2 - N_1 D_1 + N_0 D_1^2 - \frac{1}{2} N_0 D_2$$

$$N_0 = G_0(x-x')$$

$$N_1 = 3G_0(x-x')G_0^2(0) + 12G_0(x-y)G_0(0)G(y-x)$$

$$N_2 = 9G_0(x-x')G_0(0)^4 + 72G_0(x-x')G_0(y-y')G_0(0)^2 \\ + 24G_0(x-x')G_0(y-y')^4$$

$$+ 72G_0(x-y)G_0(x'-y)G_0(0)^3$$

$$+ 288G_0(x-y)G_0(x'-y)G(y-y')$$

$$+ 192G_0(x-y)G_0(y-y')G(x'-y)$$

$$+ 288G(x-y)G(y-y')G(x'-y)$$

$$D_1 = 3G_0^2(0)$$

$$D_2 = 9G_0^4(0) + 72G_0^2(y-y') + 24G_0^4(y-y')$$

$$G^{(0)} = N_0$$

$$G^{(1)} = N_1 - N_0 D_1$$

$$G^{(2)} = \frac{1}{2} N_2 - N_1 D_1 + N_0 D_1^2 - \frac{1}{2} N_0 D_2$$

Exercise #7

(pg 136)

Figure 4.2 : Greens Function

$$\begin{aligned}
 \circ - X X - \circ &\rightarrow 9 \circ \text{---} \text{---} \text{---} + 72 \circ \text{---} \text{---} \text{---} \\
 &+ 24 \circ \text{---} \text{---} \text{---} + 72 \circ \text{---} \text{---} \text{---} \\
 &+ 288 \circ \text{---} \text{---} \text{---} + 192 \circ \text{---} \text{---} \text{---} \\
 &+ 288 \circ \text{---} \text{---} \text{---}
 \end{aligned}$$

(4.12) "Greens function - modified"

$$\nabla G_0(x-x') \approx \int \frac{d^d p}{(2\pi)^d} \frac{e^{-ip(x-x')}}{p^2 + r}$$

People drew
pictures for
integrals

Others drew
pictures for
momentum in
the same function.

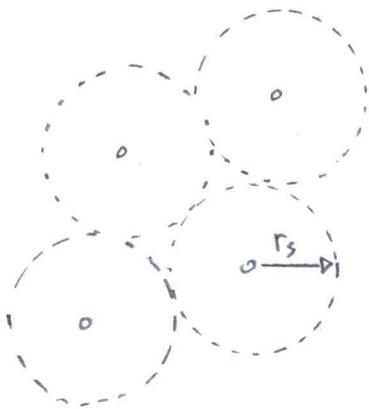
p-diagrams:

$$\begin{aligned}
 \text{Diagram with } p_1, p_2, p_3 \text{ and } -p_1, -p_2, -p_3 &\rightarrow 9 \circ \text{---} \text{---} \text{---} + 72 \circ \text{---} \text{---} \text{---} \\
 &+ 288 \circ \text{---} \text{---} \text{---} + 288 \circ \text{---} \text{---} \text{---} \\
 &+ 24 \circ \text{---} \text{---} \text{---} + 72 \circ \text{---} \text{---} \text{---}
 \end{aligned}$$

+192



Exercise #8:
(pg 192)

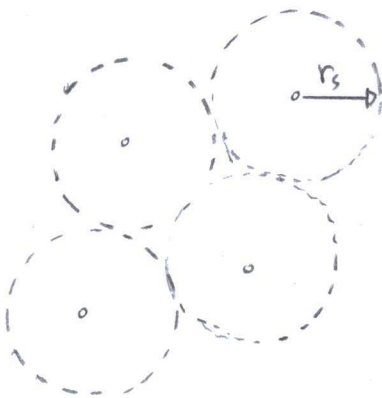


"Ball packing"

Cite: Kepler. J

"Strena seu de nive
sexangula"

Francofurti, Germany (1611)



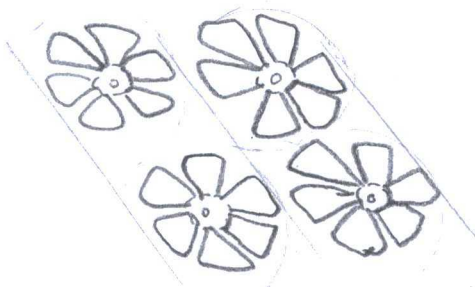
"Fermion gas"

$$r_s = \frac{\sqrt[3]{\text{Average Distance}}}{\text{Bohr radius}}$$

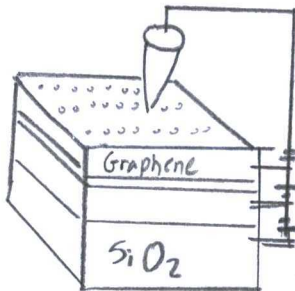
Cite: Cornell, E., Wieman E.

"Observation of Bose
Einstein Condensation
in a dilute atomic
vapor"

Colbrado (1995)



"Wigner Crystal"



Cite: Grimes, C. Adams G.

"Evidence for a
liquid-to-crystal
phase transition in
a classical, two
dimensional sheet of
Bell Labs, (1979)

Second-order expansion

(pg 189) "Free Energy"

$$F = -T \ln Z$$

$$= -T \ln \left(\int D\phi e^{-S[\phi]} \right)$$

$$= -T \ln \left(\int D\phi e^{-S_0[\phi] - S_{int}[\phi]} \right)$$

$$= -T \ln \left(\int D\phi e^{-S_0[\phi]} \left(1 - S_{int}[\phi] + \frac{1}{2} S_{int}[\phi]^2 \right) \right)$$

$$= -T \ln \left(\int D\phi e^{-S_0[\phi]} - \int D\phi e^{-S_0[\phi]} \circ S_{int}[\phi] - \int D\phi e^{-S_0[\phi]} \circ \frac{1}{2} S_{int}[\phi]^2 \right)$$

$$= -T \ln \left(Z_0 (1 - \langle S_{int}[\phi] \rangle + \frac{1}{2} \langle S_{int}^2 \rangle) \right)$$

Logarithm Expansion

$$\ln(1+x) = x - \left(\frac{1}{2}\right)x^2 + \dots$$

$$= -T \left(\underbrace{Z_0}_{F^{(0)}} - \underbrace{\langle S_{int} \rangle}_0 + \underbrace{\frac{1}{2} \langle S_{int}^2 \rangle - \frac{1}{2} \langle S_{int} \rangle_0^2}_{F^{(2)}} + \dots \right)$$

$$F^{(2)} = -\frac{T}{2} \left(\langle S_{int}^2 \rangle - \langle S_{int} \rangle^2 \right)_0$$

Disconnected Diagrams

(4.19) "Zeroth order perturbation"

$$F^{(0)} = -T \sum_{\mathbf{p}\sigma} \ln(1 + e^{-\beta(\frac{p^2}{2m} - \mu)})$$

$$\approx -\frac{2}{5} N \mu$$

(4.21) "First order perturbation"

$$F^{(1)} = \frac{T^4}{2L^3} \left\langle \sum_{\mathbf{p}\mathbf{p}'\mathbf{q}, \sigma\sigma'} \bar{\psi}_{\mathbf{p}+\mathbf{q}\sigma} \bar{\psi}_{\mathbf{p}'-\mathbf{q}\sigma'} V(\mathbf{q}) \psi_{\mathbf{p}'\sigma'} \psi_{\mathbf{p}\sigma} \right\rangle_0$$

$$= -\frac{T^2}{L^3} \sum_{\mathbf{p}\mathbf{p}'} G_{\mathbf{p}} V(\mathbf{p}-\mathbf{p}') G_{\mathbf{p}'}$$

(pg 152) "Second order perturbation"

$$F^{(2)} = -\frac{T}{2} \left(\frac{T^3}{2L^3} \right)^2 \left\langle \left(\sum_{\mathbf{p}\mathbf{p}'\mathbf{q}} \bar{\psi}_{\mathbf{p}+\mathbf{q}} \bar{\psi}_{\mathbf{p}'-\mathbf{q}} V(\mathbf{q}) \psi_{\mathbf{p}'} \psi_{\mathbf{p}} \right)^2 \right\rangle_0$$

$F^{(2)}$ is the only reliable term to $V(q=0)$ for "a wavy line" on page 190. The line indicates a zero evaluation by $V(q=0) = \frac{1}{4\pi\epsilon_0} \frac{q^2}{r^2} = 0$

Exercise #9

(pg 202)

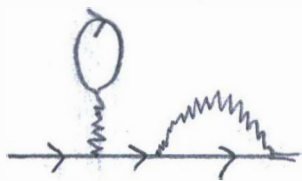
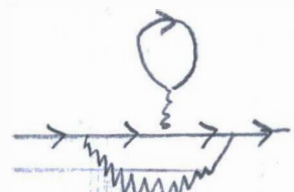
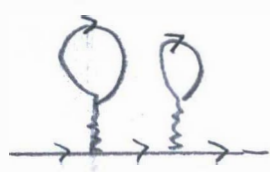
Shape	Purpose
	Green Function = $\frac{1}{ r }$
	Coefficient ($1/N$)
	Green Function difference = $\frac{1}{ r-r' }$
	positive (spin up)
	negative (spin down)

"Non-crossing approximation"

Papers challenged $O(N^2)$ degeneracy in a matrix for the computer by NoCoA.

with

1/2 N (1/2 N - 1) ...

Diagrams	Functions
	$\frac{g}{L^d} \left(\frac{1}{N} \sum G_{0,p,p'} \circ G_{0,p} \circ G_{0,p'} \circ G_{0,p+p'} \right)$
	$\frac{g}{L^d} \left(\frac{1}{N} \sum G_{0,p} \circ G_{0,p-p'} \circ G_{0,p+p'} \circ G_{0,p'} \right)$
	$\frac{g}{L^d} \left(\frac{1}{N} \sum G_{0,p} \circ G_{0,p'} \circ G_{0,p-p'} \circ G_{0,p'} \right)$

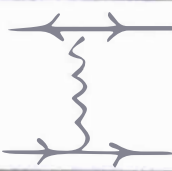
	$\frac{g}{L^d} \sum_p G_{p,0} \circ G_{0,p-p'} \circ G_{0,p+p'} \circ G_{0,p'}$
	$\frac{g}{L^d} \sum_p G_{0,p} \circ G_{0,p+p'} \circ G_{p'+p'} \circ G_{p'} \circ G_{0,p'}$
	$\frac{g}{L^d} \sum_p G_{0,p} \circ G_{0,p+p'} \circ G_{0,p'} \circ G_{0,p'+p'} \circ G_{0,p'}$

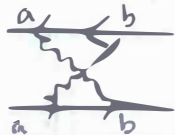
(4.44) "Non-crossing approximation"

$$\sum_p^{NCA} = -\frac{g}{L^d} \sum G_{p'}$$

∞ Unsure on the truth about efficiency against $O(N^2)$ computational operations. I bet for the d-dimensional matrix.

Exercise # 10:
(pg 205)

Schematic	Function
	"Greens Function" = $G(p)$
	Two Greens Functions $G(p)$ and $G(p')$
	Function multiplied $G(p) \circ G(p')$



$$G(p, a) \circ G(p', b) + G(p, b) \circ G(p', a)$$



$$G(p, a) \circ G(p', b) + G(p, b) \circ G(p', a) \\ + G(p, b) G(p', b)$$

Problem 4.6.1: Technical aspects of diagrammatic perturbation theory:

a) $\langle \phi(x) \phi^4(y) \phi^4(y') \phi(x') \rangle =$ Exercise #6 - pg 134

b) Exercise #6 - pg 134

c) Exercise #7 - pg 136

Problem 4.6.2: Self-consistent T-matrix approximation:

a) (4.69) "T-matrix"

$$T_n = \langle H_{imp} + H_{imp} G_{0,n}(0) H_{imp} + H_{imp} G_{0,n}(0) H_{imp} G_{0,n}(0) H_{imp} + \dots \rangle$$

where $H_{imp} = \gamma \sum \delta(r - R_i)$

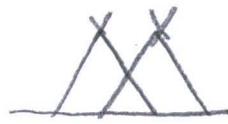
(4.22) "Green function of the electron gas"

$$G_p = \frac{1}{i\omega - \frac{p^2}{2m} + \mu} = \frac{1}{i\omega - H_0 - H_{imp}}$$

$$T_n = \langle \gamma \sum \delta(r - R_i) (1 + \gamma \sum (r - R_i) G_{0,n}(0)) \rangle$$

$$G_n = G_{0,n} + G_{0,n} < \delta (1 + \Sigma G_{0,n}) > G_{0,n}$$

b) Crossing impurity lines:



the author

showed a diagram

for difference in a Greens

function. In a one-to-one

system, a line is an

integral. Where two integrals

cancel and "leave out" an

impurity,

Problem 2 - $k_0(\epsilon) T^\pm(\epsilon) \hat{=} \hat{T}(i\omega_n \rightarrow \epsilon \pm i\delta)$ Why?

$$T^+(\epsilon) - T^-(\epsilon) = \hat{T}(i\omega_n \rightarrow \epsilon + i\delta) - \hat{T}(i\omega_n \rightarrow \epsilon - i\delta)$$

$$= \frac{1}{\gamma^{-1} + G_n^+(\epsilon)} - \frac{1}{\gamma^{-1} - G_n(\epsilon)}$$

$$= \frac{G^+(\epsilon, 0) - G^-(\epsilon, 0)}{(\gamma^{-1} + G_n^+(\epsilon))(\gamma^{-1} - G_n(\epsilon))}$$

$$= T^+ (G^+(t,0) - G^-(t,0)) T^-$$

$$= |T^+(t)|^2 v$$

Problem 4.6.3 - Kondo Effect - Perturbation :

a) (2.50) "sd-Hamiltonian"

$$\hat{H}_{sd} = \sum_{k\sigma} \epsilon_k \cdot c_{k\sigma}^\dagger c_{k\sigma} + 2J \hat{S}_d \cdot \hat{S}(0)$$

$$\text{Where } \hat{S}(0) = \sum_{k\sigma'\sigma} c_{k\sigma'}^\dagger \sigma_{\sigma\sigma'} c_{k\sigma'}$$

(from problem) "Hamiltonian"

$$\hat{H} = \sum_{k\sigma} \epsilon_k c_{k\sigma}^\dagger c_{k\sigma}$$

$$H_{imp} = 2J \hat{S} \cdot S(r=0)$$

(from problem)

$$\frac{1}{2\tau} = \pi \frac{n_{imp}}{L^d} \sum \langle | \langle k, \sigma | \hat{H}_{imp} | k', \sigma' \rangle |^2 \rangle$$

$$= \pi C_{imp} \cdot J^2 L^{-d} \sum_{\sigma'} \langle \langle \sigma | S \cdot \sigma | \sigma' \rangle \rangle$$

$$= \pi C_{imp} \cdot v J^2 S(S+1)$$

If $L=1$ and $\delta(t, \epsilon_k)=1$

also $\langle\langle \sigma | s \cdot \sigma s \cdot \sigma | \sigma \rangle\rangle$
 $= S(S+1)$

b) (from problem) "T-matrix"

$$\hat{T}^{(2)} = \hat{H}_{imp} (\epsilon^\dagger - \hat{H}_0) \hat{H}_{imp}$$

$$Re \langle k', \sigma' | \hat{T}^{(2)} | k, \sigma \rangle = Re \langle k', \sigma' | \hat{H}_{imp} | k, \sigma \rangle$$

$$= Re \langle k', \sigma' | \hat{H}_{imp} (\epsilon - H_0) \hat{H}_{imp} | k, \sigma \rangle$$

$$= Re \langle k', \sigma' | (J \sum (s \cdot \sigma) c_p^\dagger | \Omega \rangle$$

$$- J \sum (s \cdot \sigma) c_p^\dagger c_k c_p | \Omega \rangle | k, \sigma \rangle$$

$$= J^2 (s \cdot \sigma) (s \cdot \sigma) \frac{\sum \langle \Omega | c_p^\dagger c_p | \Omega \rangle}{\epsilon_k - \epsilon_p}$$

$$+ J^2 (s \cdot \sigma) (s \cdot \sigma) \frac{\sum \langle \Omega | c_p^\dagger c_k c_p^\dagger c_p c_p | \Omega \rangle}{\epsilon_k - \epsilon_p}$$

$$= J^2 \sum \frac{1}{\epsilon_k - \epsilon_p} (s \cdot \sigma)^2 [\Theta(-\xi) - \Theta(\xi)]$$

$$= J^2 \sum \frac{1}{\epsilon_k - \epsilon_p} [S(S+1) - (2n_f(\xi) - 1) S \sigma_{\sigma\sigma'}]$$

when $\Theta(\xi) = n_f(\xi)$

$\Theta(+\xi) = 1 - n_f$